

Power BI Assignment 1- Data Transformation & Data Modeling

- "List of Orders" table to only the first 500 rows

Order ID	Order Date	CustomerName	Location
B-25627	23-04-2018	Hitika	Madhya Pradesh, Indore
B-25637	26-04-2018	Ashmi	Madhya Pradesh, Indore
B-25645	01-05-2018	Yaanvi	Madhya Pradesh, Indore
B-25651	07-05-2018	Anurag	Madhya Pradesh, Indore
B-25655	11-05-2018	Nida	Madhya Pradesh, Indore
B-25663	19-05-2018	Pournamasi	Madhya Pradesh, Indore
B-25680	04-06-2018	Aayushi	Madhya Pradesh, Indore
B-25681	04-06-2018	Bhawna	Madhya Pradesh, Indore
B-25685	10-06-2018	Sheetal	Madhya Pradesh, Indore
B-25688	11-06-2018	Swetha	Madhya Pradesh, Indore
B-25690	15-06-2018	Gunjan	Madhya Pradesh, Indore
B-25692	17-06-2018	Rashmi	Madhya Pradesh, Indore
B-25701	26-06-2018	Maithilee	Madhya Pradesh, Indore
B-25703	28-06-2018	Ekta	Madhya Pradesh, Indore
B-25705	30-06-2018	Shweta	Madhya Pradesh, Indore
B-25708	01-07-2018	Kishwar	Madhya Pradesh, Indore
B-25709	01-07-2018	Aakanksha	Madhya Pradesh, Indore
B-25719	12-07-2018	Rashmi	Madhya Pradesh, Indore
B-25724	19-07-2018	Sheetal	Madhya Pradesh, Indore
B-25727	22-07-2018	Turumella	Madhya Pradesh, Indore
B-25734	29-07-2018	Pranav	Madhya Pradesh, Indore

Table: List of Orders (500 rows)

- "Order Date" column in the "List of Orders" table is set to data type 'Date'.

Name	Order Date	Format	14-03-2001 (dd-m-...
Data type	Date		Auto
Structure	Formatting		
Order ID	Order Date	CustomerName	Location
B-25627	23-04-2018	Hitika	Madhya Pradesh, Indore
B-25637	26-04-2018	Ashmi	Madhya Pradesh, Indore
B-25645	01-05-2018	Yaanvi	Madhya Pradesh, Indore
B-25651	07-05-2018	Anurag	Madhya Pradesh, Indore
B-25655	11-05-2018	Nida	Madhya Pradesh, Indore
B-25663	19-05-2018	Pournamasi	Madhya Pradesh, Indore
B-25680	04-06-2018	Aayushi	Madhya Pradesh, Indore
B-25681	04-06-2018	Bhawna	Madhya Pradesh, Indore
B-25685	10-06-2018	Sheetal	Madhya Pradesh, Indore
B-25688	11-06-2018	Swetha	Madhya Pradesh, Indore

- Data type of “Amount” and “Target” columns to ‘Fixed Decimal Number’.

Amount

Name: Amount

Data type: Fixed decimal num...

Format: Currency

Structure: [X] [✓]

Order ID	Amount	Profit	Quantity	Category	Sub-Category
B-25603	£12	1	2	Clothing	Hank
B-25608	£257	23	5	Clothing	Hank
B-25615	£68	20	5	Clothing	Hank
B-25616	£42	12	5	Clothing	Hank
B-25624	£26	12	3	Clothing	Hank
B-25625	£97	29	2	Clothing	Hank
B-25635	£40	16	3	Clothing	Hank
B-25638	£154	39	3	Clothing	Hank
B-25654	£34	12	3	Clothing	Hank
B-25656	£6	3	1	Clothing	Hank

Target

Name: Target

Data type: Fixed decimal num...

Format: Currency

Structure: [X] [✓]

Month of Order Date	Category	Target
01 April 2018	Clothing	£12,000
01 May 2018	Clothing	£12,000
01 June 2018	Clothing	£12,000
01 July 2018	Clothing	£14,000
01 August 2018	Clothing	£14,000
01 September 2018	Clothing	£14,000
01 October 2018	Clothing	£16,000
01 November 2018	Clothing	£16,000
01 December 2018	Clothing	£16,000
01 January 2019	Clothing	£16,000
01 February 2019	Clothing	£16,000
01 March 2019	Clothing	£16,000

- Format the "CustomerName" column into proper case, ensuring consistent capitalization for each word.

Date	CustomerName	
April 2018	Hitika	Mac
April 2018	Ashmi	Mac
1 May 2018	Yaanvi	Mac
7 May 2018	Anurag	Mac
1 May 2018	Nida	Mac
9 May 2018	Pournamasi	Mac
1 June 2018	Aayushi	Mac
1 June 2018	Bhawna	Mac
1 June 2018	Sheetal	Mac
June 2018	Swetha	Mac
1 June 2018	Gunjan	Mac
1 June 2018	Rashmi	Mac
1 June 2018	Maithilee	Mac
1 June 2018	Ekta	Mac
1 June 2018	Shweta	Mac
1 July 2018	Kishwar	Mac
1 July 2018	Aakanksha	Mac
2 July 2018	Rashmi	Mac
9 July 2018	Sheetal	Mac
2 July 2018	Turumella	Mac
9 July 2018	Pranav	Mac
Column: CustomerName (332 distinct)		

- the format 'City, State'.

Order ID	Order Date	CustomerName	Location
B-25627	23 April 2018	Hitika	Madhya Pradesh, Indore
B-25637	26 April 2018	Ashmi	Madhya Pradesh, Indore
B-25645	01 May 2018	Yaanvi	Madhya Pradesh, Indore
B-25651	07 May 2018	Anurag	Madhya Pradesh, Indore
B-25655	11 May 2018	Nida	Madhya Pradesh, Indore
B-25663	19 May 2018	Pournamasi	Madhya Pradesh, Indore
B-25680	04 June 2018	Aayushi	Madhya Pradesh, Indore
B-25681	04 June 2018	Bhawna	Madhya Pradesh, Indore
B-25685	10 June 2018	Sheetal	Madhya Pradesh, Indore
B-25688	11 June 2018	Swetha	Madhya Pradesh, Indore
B-25690	15 June 2018	Gunjan	Madhya Pradesh, Indore
B-25692	17 June 2018	Rashmi	Madhya Pradesh, Indore
B-25701	26 June 2018	Maithilee	Madhya Pradesh, Indore
B-25703	28 June 2018	Ekta	Madhya Pradesh, Indore
B-25705	30 June 2018	Shweta	Madhya Pradesh, Indore
B-25708	01 July 2018	Kishwar	Madhya Pradesh, Indore
B-25709	01 July 2018	Aakanksha	Madhya Pradesh, Indore
B-25719	12 July 2018	Rashmi	Madhya Pradesh, Indore
B-25724	19 July 2018	Sheetal	Madhya Pradesh, Indore
B-25727	22 July 2018	Turumella	Madhya Pradesh, Indore
B-25734	29 July 2018	Pranav	Madhya Pradesh, Indore

- divided by "Amount".

X								Data	
Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	Search	
B-25603	£12	1	2	Clothing	Hankerchief	8	Profit	List of Orders	
B-25608	£257	23	5	Clothing	Hankerchief	9	Profit	CustomerName	
B-25615	£68	20	5	Clothing	Hankerchief	29	Profit	Location	
B-25616	£42	12	5	Clothing	Hankerchief	29	Profit	Order Date	
B-25624	£26	12	3	Clothing	Hankerchief	46	Profit	Order ID	
B-25625	£97	29	2	Clothing	Hankerchief	30	Profit	Order Details	
B-25635	£40	16	3	Clothing	Hankerchief	40	Profit	Amount	
B-25638	£154	39	3	Clothing	Hankerchief	25	Profit	Category	
B-25654	£34	12	3	Clothing	Hankerchief	35	Profit	Order ID	
B-25656	£6	3	1	Clothing	Hankerchief	50	Profit	Profit	
B-25656	£56	18	2	Clothing	Hankerchief	32	Profit	Profit Margin	
B-25670	£24	1	2	Clothing	Hankerchief	4	Profit	Profit Status	
B-25670	£14	2	1	Clothing	Hankerchief	14	Profit	Quantity	
B-25730	£18	8	2	Clothing	Hankerchief	44	Profit	Sub-Category	
B-25751	£32	7	3	Clothing	Hankerchief	22	Profit	Order Details (2)	
B-25757	£106	15	7	Clothing	Hankerchief	14	Profit	Orders Data	
B-25757	£14	5	1	Clothing	Hankerchief	36	Profit	Sales target	
B-25757	£17	7	3	Clothing	Hankerchief	41	Profit		
B-25771	£148	59	3	Clothing	Hankerchief	40	Profit		
B-25783	£26	2	2	Clothing	Hankerchief	8	Profit		
B-25784	£15	4	1	Clothing	Hankerchief	27	Profit		

able: Order Details (1,500 rows)

Update available (click to d

- Add a new conditional column named "Profit Status" based on the values in the "Profit" column. The conditions are as follows: if the profit is less than 0, the label should be "Loss"; if the profit equals 0, the label should be "Break-Even"; and if the profit is greater than 0, the label should be "Profit".

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status
B-25603	£12	1	2	Clothing	Hankerchief	8	Profit
B-25608	£257	23	5	Clothing	Hankerchief	9	Profit
B-25615	£68	20	5	Clothing	Hankerchief	29	Profit
B-25616	£42	12	5	Clothing	Hankerchief	29	Profit
B-25624	£26	12	3	Clothing	Hankerchief	46	Profit
B-25625	£97	29	2	Clothing	Hankerchief	30	Profit
B-25635	£40	16	3	Clothing	Hankerchief	40	Profit
B-25638	£154	39	3	Clothing	Hankerchief	25	Profit
B-25654	£34	12	3	Clothing	Hankerchief	35	Profit
B-25656	£6	3	1	Clothing	Hankerchief	50	Profit
B-25656	£56	18	2	Clothing	Hankerchief	32	Profit
B-25670	£24	1	2	Clothing	Hankerchief	4	Profit
B-25670	£14	2	1	Clothing	Hankerchief	14	Profit
B-25730	£18	8	2	Clothing	Hankerchief	44	Profit
B-25751	£32	7	3	Clothing	Hankerchief	22	Profit
B-25757	£106	15	7	Clothing	Hankerchief	14	Profit
B-25757	£14	5	1	Clothing	Hankerchief	36	Profit
B-25757	£17	7	3	Clothing	Hankerchief	41	Profit
B-25771	£148	59	3	Clothing	Hankerchief	40	Profit
B-25783	£26	2	2	Clothing	Hankerchief	8	Profit
B-25784	£15	4	1	Clothing	Hankerchief	27	Profit

Merging Data (Joins):

- Merge the "List of Orders" and "Order Details" tables into a new single table named "Orders Data" based on the "Order ID" relationship.

Order ID	Order Date	CustomerName	Location	Order Details.Amount	Order Details.Profit	Order Details.Quantity	Order Details.Category
B-26099	30 March 2019	Bhishm	Maharashtra, Mumbai	£207	37	4	Clothing
B-26097	28 March 2019	Vini	Karnataka, Bangalore	£14	5	1	Clothing
B-26097	28 March 2019	Vini	Karnataka, Bangalore	£19	8	2	Clothing
B-26097	28 March 2019	Vini	Karnataka, Bangalore	£97	12	2	Clothing
B-26091	27 March 2019	Manju	Andhra Pradesh, Hyderabad	£29	10	4	Clothing
B-26093	27 March 2019	Sarita	Maharashtra, Pune	£49	5	4	Clothing
B-26086	26 March 2019	Mukesh	Haryana, Chandigarh	£25	2	2	Clothing
B-26088	26 March 2019	Bhavna	Sikkim, Gangtok	£11	5	2	Clothing
B-26089	26 March 2019	Kanak	Goa, Goa	£27	4	1	Clothing
B-26087	26 March 2019	Vandana	Himachal Pradesh, Simla	£46	13	3	Clothing
B-26077	22 March 2019	Divsha	Rajasthan, Jaipur	£62	11	7	Clothing
B-26079	22 March 2019	Hazel	Karnataka, Bangalore	£18	3	2	Clothing
B-26081	22 March 2019	Aarushi	Tamil Nadu, Chennai	£24	11	3	Clothing
B-26073	21 March 2019	Pournamasi	Madhya Pradesh, Indore	£37	17	3	Clothing
B-26073	21 March 2019	Pournamasi	Madhya Pradesh, Indore	£122	11	4	Clothing
B-26071	19 March 2019	Subhashree	Jammu and Kashmir, Kashmir	£21	4	3	Clothing
B-26070	18 March 2019	Shruti	Karnataka, Bangalore	£14	7	2	Clothing
B-26064	16 March 2019	Ankita	Maharashtra, Mumbai	£61	3	4	Clothing
B-26067	16 March 2019	Tulika	Madhya Pradesh, Bhopal	£53	2	4	Clothing
B-26058	11 March 2019	Aditya	Punjab, Chandigarh	£212	97	7	Clothing
B-26054	10 March 2019	Megha	Maharashtra, Mumbai	£146	98	6	Clothing

Orders Data (1,500 rows)

Update available (d

Handling Missing Data & Duplicate Data:

- Identify missing values in the data and determine a strategy to address them.
- Check for duplicate rows and define a strategy to handle duplicates.

Sorting and Filtering Data:

- In the 'Orders Data' table, utilize sorting and filtering techniques on columns like

Order Date, State or Category to analyze data based on specific criteria:

- ◆ Sort the orders by Order Date in descending order to analyze recent trends.

Formula: `= Table.Sort("#Expanded Order Details",{{"Order Date", order.Descending}})`

Order ID	Order Date	CustomerName	Location	Order Details.Amount
1 B-26100	31-03-2019	Hitika	Madhya Pradesh, Indore	
2 B-26100	31-03-2019	Hitika	Madhya Pradesh, Indore	
3 B-26100	31-03-2019	Hitika	Madhya Pradesh, Indore	
4 B-26099	30-03-2019	Bhishm	Maharashtra, Mumbai	
5 B-26099	30-03-2019	Bhishm	Maharashtra, Mumbai	2,
6 B-26099	30-03-2019	Bhishm	Maharashtra, Mumbai	
7 B-26099	30-03-2019	Bhishm	Maharashtra, Mumbai	
8 B-26098	29-03-2019	Pinky	Jammu and Kashmir, Kashmir	
9 B-26098	29-03-2019	Pinky	Jammu and Kashmir, Kashmir	
10 B-26098	29-03-2019	Pinky	Jammu and Kashmir, Kashmir	
11 B-26098	29-03-2019	Pinky	Jammu and Kashmir, Kashmir	
12 B-26098	29-03-2019	Pinky	Jammu and Kashmir, Kashmir	
13 B-26098	29-03-2019	Pinky	Jammu and Kashmir, Kashmir	
14 B-26096	28-03-2019	Atharv	West Bengal, Kolkata	
15 B-26097	28-03-2019	Vini	Karnataka, Bangalore	
16 B-26097	28-03-2019	Vini	Karnataka, Bangalore	
17 B-26096	28-03-2019	Atharv	West Bengal, Kolkata	
18 B-26097	28-03-2019	Vini	Karnataka, Bangalore	
19 B-26096	28-03-2019	Atharv	West Bengal, Kolkata	
20 B-26096	28-03-2019	Atharv	West Bengal, Kolkata	
21 B-26097	28-03-2019	Vini	Karnataka, Bangalore	

Query Settings

PROPERTIES

Name: Orders Data

APPLIED STEPS

Source, Expanded Order Details, Sorted Rows

- ◆ Filter the orders to focus only on a specific state (e.g., Tamil Nadu) for

regional analysis.

Formula: `= Table.SelectRows("#Changed Type", each ([Category] = "Clothing"))`

Month of Order Date	Category	Target
01-04-2018	Clothing	12,000.00
01-05-2018	Clothing	12,000.00
01-06-2018	Clothing	12,000.00
01-07-2018	Clothing	14,000.00
01-08-2018	Clothing	14,000.00
01-09-2018	Clothing	14,000.00
01-10-2018	Clothing	16,000.00
01-11-2018	Clothing	16,000.00
01-12-2018	Clothing	16,000.00
01-01-2019	Clothing	16,000.00
01-02-2019	Clothing	16,000.00
01-03-2019	Clothing	16,000.00

Query Settings

PROPERTIES

Name: Sales target

APPLIED STEPS

Source, Promoted Headers, Changed Type, Filtered Rows

- Duplicate the “Order Details” table and calculate the count of each Order ID, average profit by Category or total amount by Sub-Category.

- profit by Category or total amount by Sub-Category.

Queries [6]

List of Orders

Order Details

Sales target

Orders Data

Order Details (2)

Sales target (2)

fx

= Table.Group("#Changed Type2", {"Table ID"}, {"Count", each Table.RowCount(_), Int64.Type})

	Order ID	Count
1	B-25601	4
2	B-25602	5
3	B-25603	8
4	B-25604	2
5	B-25605	1
6	B-25606	1
7	B-25607	1
8	B-25608	4
9	B-25609	2
10	B-25610	6
11	B-25611	1
12	B-25612	1
13	B-25613	1
14	B-25614	2
15	B-25615	1
16	B-25616	4
17	B-25617	1
18	B-25618	2
19	B-25619	1
20	B-25620	1
21	B-25621	3
22	B-25622	1

Query Settings

PROPERTIES

Name

Order Details (2)

All Properties

APPLIED STEPS

Source

Promoted Headers

Changed Type

Added Custom

Changed Type1

Added Conditional Column

Changed Type2

Grouped Rows

- Order Date.

Queries [6]

List of Orders

Order Details

Sales target

Orders Data

Order Details (2)

Sales target (2)

Month of Order Date

1.2 Total target amount

1	01-04-2018	31400
2	01-05-2018	31500
3	01-06-2018	31600
4	01-07-2018	33800
5	01-08-2018	33900
6	01-09-2018	34000
7	01-10-2018	36100
8	01-11-2018	36300
9	01-12-2018	36400
10	01-01-2019	43500
11	01-02-2019	43600
12	01-03-2019	43800

= Table.Group(#"Changed Type", {"Month of Order Date"}, {{ "Total target amount", each List.Sum(

Query Settings

PROPERTIES

Name

Sales target (2)

All Properties

APPLIED STEPS

Source

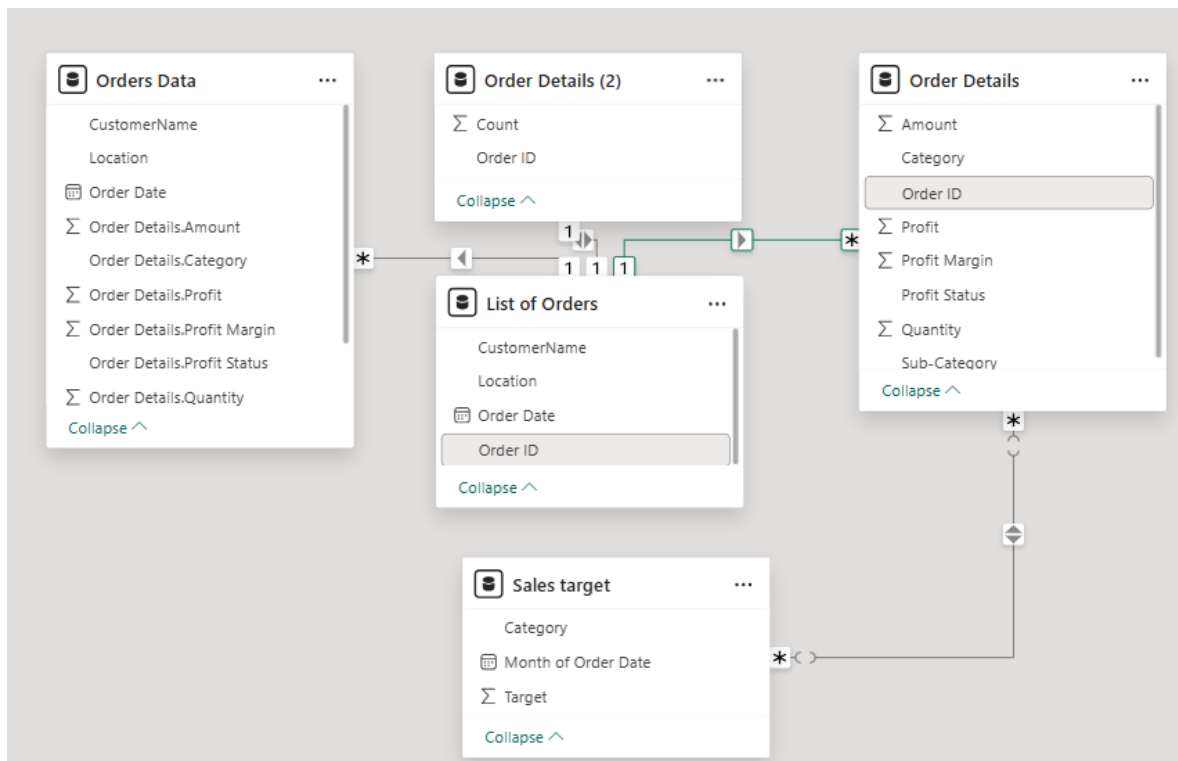
Promoted Headers

Changed Type

Grouped Rows

- Establish a relationship between the “List of Orders” and “Order Details” tables using the ‘Order ID’ column.

- Establish a relationship between the “List of Orders” and “Order Details” tables using the ‘Order ID’ column.



- Build a relationship between the “Order Details” and “Sales Target” tables based on the ‘Category’ column. Click "Manage relationships" and ensure this relationship is active.

